

Yongfei Deng

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Education

Doctor of Philosophy (Ph.D.) in Physical Oceanography

Southern University of Science and Technology, Shenzhen, China

2019.09 – 2023.06

Thesis: “The Interannual Variability and Driving Mechanism of Shelf Circulation and Cross-Shelf Water Exchange in the Northern South China Sea”

Courses: Numerical Simulation of Ocean Circulation; Physical Oceanography; Advanced Numerical Methods; Advanced Continuum Mechanics; Academic Writing and Presentation; Global Hydrology and Frontier of Environmental Science

Master of Science (M.Sc.) in Marine Science

Shanghai Ocean University, Shanghai, China

2016.09 – 2019.06

Thesis: “The Variabilities, Impacts, and Driving Mechanisms of Bering Strait Throughflow”

Honors: Excellent Thesis

Bachelor of Science (B.Sc.) in Resources, Environment, and Urban-Rural Planning Management

Guangdong University of Finance and Economics, Guangzhou, China

2012.08 – 2016.06

Thesis: “The Climate Responses to the El Niño Events from 2000 in Guangdong Province, China”

Honors: Outstanding Graduate; Excellent Thesis

Research Experience

Postdoctoral Research Scholar

North Carolina State University, Raleigh, NC, USA

2024.10 – Present

- Conducting advanced research in physical oceanography with a focus on data assimilation techniques to enhance ocean model accuracy.
- Using numerical model and ML/AI to study eddies in Gulf of Mexico and water mass transformation in Caribbean Sea.

Visiting Researcher and Research Associate

University of New South Wales, Canberra, Australia

2022.09 – 2024.10

- Conducted numerical modeling studies on sediment and physical dynamics under multiple stressors in Batemans Bay, NSW.
- Analyzed the impacts of regional climate change and infrastructure construction on sediment transport and hydrodynamics.
- Identified physical mechanisms governing mass exchange between semi-enclosed bays and adjacent shelves.

Ph.D. Researcher

Southern University of Science and Technology, Shenzhen, China

2019.09 – 2023.06

- Investigated low-frequency variability of shelf circulation and cross-shelf water exchange in the Northern South China Sea using ROMS modeling and observational data.
- Studied the role of coastally-trapped waves in modulating water exchange in the Taiwan Strait.

Master's Researcher

Shanghai Ocean University, Shanghai, China

2016.04 – 2019.05

- Explored air-ice-sea dynamics in the Arctic Ocean using in situ data and FVCOM model output.
- Focused on key regions such as the Bering Strait, Chukchi Sea, Beaufort Sea, and Canadian Arctic Archipelago.

Research Interests

- Physical oceanography, focusing on multi-scale ocean dynamics, air-sea-ice interactions, and cross-shelf water exchange.
- Numerical modeling and data assimilation to improve predictions of ocean circulation and climate dynamics.
- Investigating the impacts of climate change on coastal and polar ocean systems.

Peer-Reviewed Publications

1. **Deng, Y.**, T. Wu, J. Gradone, T. Miles, D. Wilson, R. He: ENSO-Driven Modulation of the Caribbean Subsurface Salinity Maximum. *Communications Earth & Environment*. (submitted; preprint available on EarthArXiv)
2. **Deng, Y.**, T. Wu, R. He: Tropical cyclones intensify mesoscale eddy variability and accelerate Western Boundary Current instability. *Communications Earth & Environment*. (submitted; preprint available on EarthArXiv)
3. Mirabito, C., P. J. Haley, E. M. Mule, A. V. Rodriguez, S. L. Morey, E. P. Chassignet, S. M. Glenn, T. N. Miles, D. Aragon, K. Coleman, M. Smith, S. F. DiMarco, S. Mahmud, X. Ge, A. H. Knap, B. Jaimes de la Cruz, L. K. Shay, M. Leber, R. Ramos, H. Nowak, J. Storie, A. Romer, M. Tenreiro, E. Pallàs-Sanz, J. Sheinbaum, P. Pérez-Brunius, R. He, **Y. Deng**, T. Wu, A. Bower, H. H. Furey, K. A. Donohue, J. van Smirren, P. Hogan, G. Jacobs, M. Feldman, F. K. Wiese, M. Khadka, and P. F. J. Lermusiaux, 2025: Real-Time Optimal Planning and Adaptive Sampling for Multi-Platform Operations in the Gulf of Mexico. *OCEANS 2025 - Great Lakes*, 1–10.
4. **Deng, Y.**, H. Wang, X. Zhou, G. Yang, 2026: Estuarine circulation and bay-shelf exchange under river and wave forcing in Batemans Bay, New South Wales, Australia. *Ocean Modelling*. (under review)
5. **Deng, Y.**, H. Wang, Y. Wang, Z. Li, G. Yang, 2026: High-Frequency Water-Level Oscillations in a Semi-Enclosed Embayment. *Water*, 18, 5, 611.
6. **Deng, Y.**, Z. Liu, et al., 2022: Climatic Controls on the Interannual Variability of Shelf Circulation in the Northern South China Sea. *Journal of Geophysical Research: Oceans*, 127(7).
7. Liu, Z., **Y. Deng**, et al., 2021: Progress on Circulation Dynamics in the East China Sea and Southern Yellow Sea. *Progress in Oceanography*, 193, 102553.
8. Liu, Z., **Y. Deng**, et al., 2021: Progress of studies on circulation dynamics in the East China Sea: The Kuroshio exchanges with the shelf currents. *Frontiers in Marine Science*, 8, 620910.

9. Liu, Z., J. Gan, J. Hu, H. Wu, Z. Cai and **Y. Deng**, 2021: Advances on coastal and estuarine circulations around the Changjiang Estuary in the recent decades (2000–2020). *Frontiers in Marine Science*, 8, 615929.
10. Zhang, Y., **Y. Deng**, and C. Chen, 2020: Main Characteristics and Influence Factors of the Pacific Inflow in the Region of Bering Strait: A Review. *Chinese Journal of Polar Research*, 32(2), 151–163. (In Chinese with English abstract)
11. **Deng, Y.**, Gao, G., Zhang, Y. & Chen, C., 2019: Seasonal and interannual variability of Bering Strait Throughflow from AO-FVCOM and observation. *Journal of Ocean University of China*, 18(3), 615–625.
12. **Deng, Y.**, Gao, G., Zhang, Y. & Chen, C., 2019: The temporal-spatial variability of Chukchi sea ice area under the influence of Bering Strait Throughflow. *Chinese Journal of Polar Research*, 31(4), 383–392. (In Chinese with English abstract)

Conferences & Workshops

- AGU Fall Conference, New Orleans, LN, United States, December 2025 (poster presentation)
- UGOS meeting, WHOI, MA, United States, September 2025 (Talk: CNAPS2-From short-term forecast to long-term climate variability analysis; Posters presentation).
- 2nd Coastal Resilience Workshop, UNSW Canberra, ACT, April 2024.
- 1st Coastal Resilience Workshop, UNSW Canberra, ACT, November 2023.
- 6th Conference on Earth Science, Tongji University, Shanghai, July 2021 (Poster presentation).
- Second China Polar Youth Forum, University of Science and Technology of China, May 2019 (Poster presentation; First Prize).
- China Symposium on Polar Science, Xiamen University, October 2018 (Poster presentation).

Teaching Experience

Casual Academic

School of Science, UNSW Canberra, Australia

2023.02 – 2024.09

- Demonstrated lab experiments and graded students' lab reports in two oceanographic courses: *Ocean Circulation and Mixing* and *Australian Waters*.
- Assisted in conducting tasks and surveys during the *Ocean Field School* at Batemans Bay in April 2023 and 2024.
- Designed and demonstrated lab sessions and quizzes for the summer school program, *Introduction to Oceanography*.

Fieldwork Experience

- Sino-Norway Field Measurements, Greenland-Iceland-Norwegian (Nordic) Seas, May-July 2017 (Operation of CTD and Aanderaa Current Meter).
- Ocean Field School, Batemans Bay, Australia, April 2023 (Operation of tide and wave gauges, ADCP).
- Ocean Field School, Batemans Bay, Australia, April 2024 (Operation of tide and wave gauges, ADCP).

Skills

- **Programming:** Python, MATLAB, Fortran, NCL, LaTeX
- **Systems:** Linux/Unix, Shell, HPC
- **Modeling:** ROMS, ENKF-c, FVCOM, AI/ML/DL
- **Languages:** English (Fluent), Cantonese (Native), Mandarin (Native)

Referees

1. Ruoying He, Ph.D.

Professor, School of Science, North Carolina State University, USA

Email: rhe@ncsu.edu

2. Xiaohua Wang, Ph.D.

Professor, School of Science, UNSW Canberra, Australia

Email: x.h.wang@unsw.edu.au

3. Guoping Gao, Ph.D.

Professor, College of Marine Ecology and Environment, Shanghai Ocean University, China

Email: gpgao@shou.edu.cn

4. Qi Quan, Ph.D.

Associate Professor, State Key Laboratory of Estuarine and Coastal Research, East China Normal University, China

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